FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

# CS0065
# (INTELLIGENT SYSTEMS)

## TECHNICAL ASSESSMENT

# 2

## RULE – BASED AGENT

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## I. BACKGROUND INFORMATION

A Rule-Based Agent is a type of intelligent agent that operates by following predefined rules to decide its actions. In this lab, students will work with a vacuum cleaner agent that can clean rooms or move between rooms depending on their state (dirty or clean).

The goal is to understand how agents perceive the environment, act according to rules, and update the environment.

## II. TECHNICAL ASSESSMENT

### RULE – BASED AGENT

***Objective***

At the end of this laboratory activity, students are expected to:

1. Explain the concept of rule-based agents.
2. Implement a rule-based agent in Python.
3. Interact with an environment by perceiving and acting based on rules.
4. Observe and analyze the behavior of an agent in a controlled environment.

**Materials Needed**

- Computer with Python 3.x installed
- Text editor or IDE (e.g., VSCode, PyCharm)
- Optional: Jupyter Notebook for interactive testing

**General Instructions**

1. This **Technical Assessment (TA2)** must be completed individually.
2. Follow the Task below.
3. Include clear comments in your code to describe its logic.
4. Each program should display the correct input, process, and output.
5. Ensure that your script displays clear and readable outputs.
6. Compile all Python codes and corresponding outputs into one PDF file.
7. Save it as **TA2_lastname.pdf**
   *(Example: TA2_Patilano.pdf)*
8. Submit the **python file** and **PDF** file.

**Laboratory Task**

**Pre-Lab Preparation**

1. Review the Python concepts:
   - Classes and objects
   - If-Else statements
   - Loops (for loop)

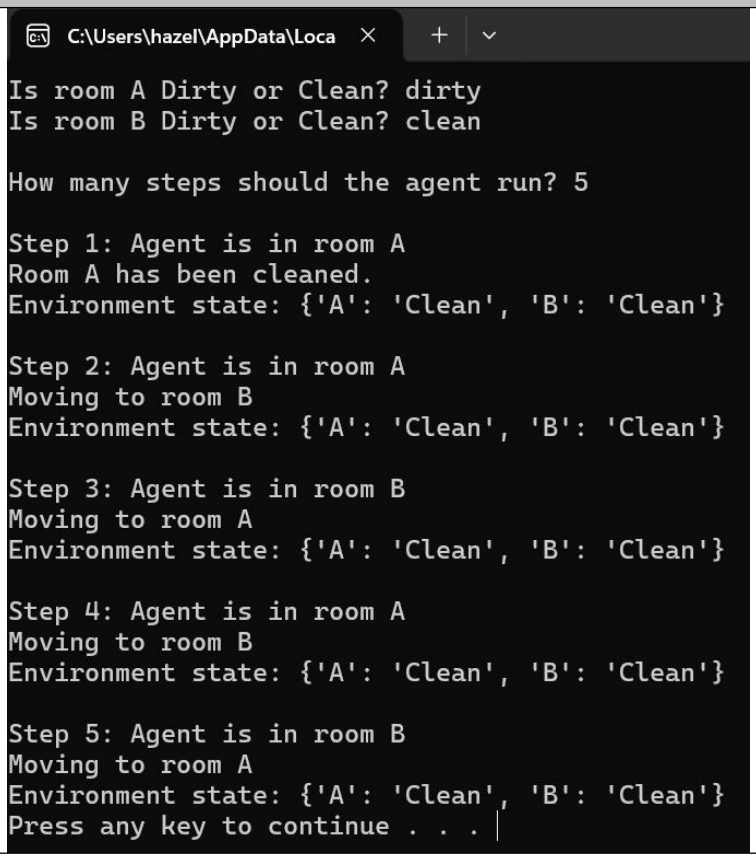
- Functions / methods

2. Understand the environment:
   - Two rooms: A and B
   - Each room can be Dirty or Clean

**Task 1: Set Up the Environment**

1. Initialize two rooms using a Python dictionary.
2. Allow the user to set the **initial state** of each room (dirty or clean).
3. Write a function to **check if a room is dirty** and a function to **clean a room**.

Sample Output




**Task 2: Implement the Rule-Based Agent**

1. Create a class for the agent.
2. Implement the **perceive-and-act method**:
   - Rule 1: If the current room is dirty, clean it.
   - Rule 2: If the current room is clean, move to the other room.
3. Implement a **move method** to switch rooms.

**Task 3: Run the Simulation**

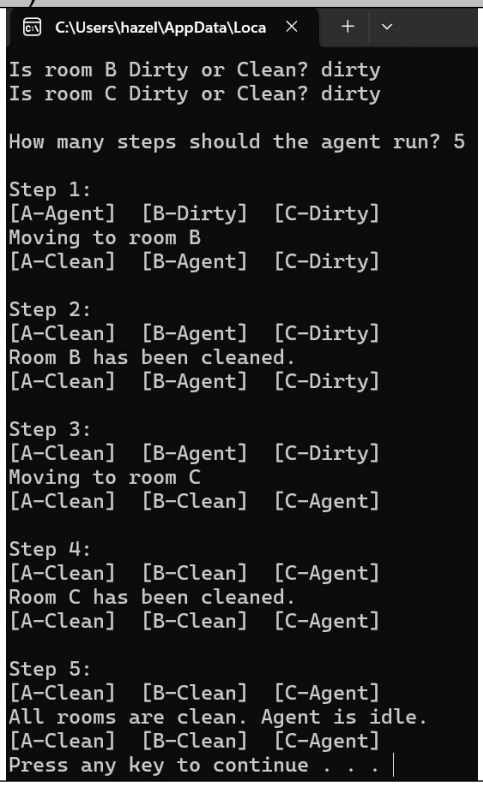
1. Ask the user how many steps the simulation should run.
2. Run the agent in a loop for the specified steps.
3. After each step, print:

- The agent's current location
- The action taken (cleaned/moved)
- Current state of the environment

### Optional (Bonus Task)

- Add a **third room (C)** and update the agent logic.
- Make the agent **randomly choose which dirty room to clean next**.
- Visualize the agent moving between rooms using a **text-based grid**.

Sample Output (Bonus Task)




### Your Code - Laboratory Tasks

```
def check_room(room, rooms):
    print(f"Agent checks Room {room}.")
    return rooms[room] == "Dirty"
```

```
def clean_room(room, rooms):
    print(f"Agent cleans Room {room}.")
    rooms[room] = "Clean"
```

```
def agent(rooms):
    current_room = "A"

    while True:
        if check_room(current_room, rooms):
            clean_room(current_room, rooms)
        else:
            print(f"Room {current_room} is already clean.")

        # Stop once both rooms are clean
        if all(state == "Clean" for state in rooms.values()):
            print("Both rooms are clean. Agent stops.")
            break

        # Move to the other room
        if current_room == "A":
            current_room = "B"
        else:
            current_room = "A"

        print(f"Agent moves to Room {current_room}.")


def get_initial_state(room):
    while True:
        state = input(
            f"Enter initial state of Room {room} (Dirty/Clean): "
        ).strip().capitalize()

        if state in ("Dirty", "Clean"):
            return state

        print("Invalid input. Please enter Dirty or Clean.")


# Get initial room states from the user
rooms = {
    "A": get_initial_state("A"),
    "B": get_initial_state("B")
}

print("\nInitial room states:")
for room, state in rooms.items():
    print(f"Room {room}: {state}")
```

```
print("\nStarting agent simulation...\n")
agent(rooms)
```

```
print("\nFinal room states:")
for room, state in rooms.items():
    print(f"Room {room}: {state}")
```

**Task 4: Analysis Questions**

After running the simulation, answer the following:

1. What rules does the agent follow? The rules the agent follows is if the room is clean, the agent moves to the other room. But if it's dirty, the agent will clean the room before moving to the other room.
2. How does the agent decide when to clean and when to move? The agent decides when to move only if the current room is clean. The agent decides to clean the room if it checks the room and finds that it is dirty based on the state.
3. What happens if both rooms are already clean at the start? If both rooms are already clean, the agent won't move since it already knows the state of both rooms from the dictionary in the program. It's only programmed to move if it knows there's a dirty room.
4. How would you modify the agent to handle **more than two rooms**? In order to handle more than two rooms, a while loop that will keep the agent moving until all room states in the dictionary are set to "clean" will work. It will only stop if all the rooms are clean.